Department of Mathematics, IIT Madras

MA 2102 - Differential Equations, ODD Semester, 2026-2027

Assignment-7 : Sturm-Liouville Problems

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1. Write the parametric Bessel equation, Legendre equation and Chebyshev equation in Sturm-Liouville form with suitable choice of $p(x)$, $q(x)$ and $r(x)$.

2. Show that $y''+fy'+(g+\lambda h)y = 0$ can be transformed into Sturm-Liouville form if $p = \exp\left(\int f\,dx\right)$, $q = pg$ and $r = hp$.

3. Find the eigenvalues and eigenfunctions of the following Sturm-Liouville problems:

   (a) $y'' + \lambda y = 0;\ y(0) = 0,\ y'(\pi) = 0.$

   (b) $y'' + 4y' - \lambda y = 0;\ y'(0) = 0,\ y'(2) = 0.$

   (c) $y'' + \lambda y = 0;\ y(0) = y(2\pi),\ y'(0) = y'(2\pi).$

   (d) $y'' + 2y' + (\lambda + 1)y = 0;\ y(0) = 0,\ y(\pi) = 0.$

   (e) $y'' - y' + \lambda y = 0;\ y(0) = y(L) = 0.$

   (f) $(xy')' + (\lambda/x)y = 0;\ y(1) = 0,\ y'(e) = 0.$

   (g) $(e^{2x}y')' + e^{-2x}(\lambda + 1)y = 0;\ y(0) = 0,\ y(\pi) = 0.$

   (h) $4(e^{-x}y')' + (1 + \lambda)e^{x}y = 0;\ y(0) = 0,\ y(1) = 0.$

   (i) $\left((x^2 + 1)y'\right)' + \dfrac{\lambda}{x^2 + 1}y = 0;\ y'(0) = 0 = y'(1).$

   (j) $x^2y'' - 3xy' + \lambda y = 0;\ y'(0) = 0,\ y'(\pi) = 0.$

   (k) $x^2y'' + 4xy' + 4\lambda y = 0;\ y'(0) = 0,\ y(2) = 0.$

   (l) $x^2y'' + xy' + \lambda y = 0;\ y(1) = y(e) = 0.$

   (m) $x^2y'' - xy' + (\lambda + 1)y = 0;\ y(1) = 0,\ y(e) = 0.$

4. Write down the conclusion of the orthogonality condition for eigenfunctions for the following SLP:

   (a) $y'' + \lambda y = 0;\ y(0) = 0,\ y'(1) = 0.$

   (b) $y'' + \lambda y = 0;\ y'(0) = 0,\ y(1) + y'(1) = 0.$

   (c) $y'' + \lambda y = 0;\ y(0) = 0,\ y(L) = 0.$

   (d) $x^2y'' + xy' + (\lambda^2x^2 - m^2)y = 0,\ 0 < x < a;\ y(0)$ is finite, $y(a) = 0.$

5. Find the eigenfunction expansions $\sum_n a_n y_n$ for the following functions by using the orthogonal eigenfunctions of the problem 4(a): (a) $f(x) = 1$, $0 \le x \le 1$; (b) $f(x) = x$, $0 \le x \le 1$; (c) $f(x) = 1 - x$, $0 \le x \le 1$;

(c) $f(x) = \begin{cases} 1, & 0 \le x < \dfrac{1}{2}, \\ 0, & \dfrac{1}{2} \le x \le 1; \end{cases}$ (d) $f(x) = \begin{cases} 2x, & 0 \le x < \dfrac{1}{2}, \\ 1, & \dfrac{1}{2} \le x \le 1. \end{cases}$

6. Find the orthonormal eigenfunction expansions $\sum_n a_n y_n$ for the following functions by using the normalized eigenfunctions of the problem 4(b): (a) $f(x) = 1$, $0 \le x \le 1$; (b) $f(x) = x$, $0 \le x \le 1$; (c) $f(x) = 1 - x$, $0 \le x \le 1$;

(c) $f(x) = \begin{cases} 1, & 0 \le x < \dfrac{1}{2}, \\ 0, & \dfrac{1}{2} \le x \le 1; \end{cases}$ (d) $f(x) = \begin{cases} 2x, & 0 \le x < \dfrac{1}{2}, \\ 1, & \dfrac{1}{2} \le x \le 1. \end{cases}$

7. Define the Sturm-Liouville operator L as

$$L = \frac{d}{dx}\left(p\,\frac{d}{dx}\right) + q.$$

Then, prove the following two identities:

(a) $uLv - vLu = \big[p(uv' - u'v)\big]'$ (Lagrange's Identity)

(b) $\displaystyle\int_a^b (uLv - vLu)\,dx = \big[p(uv' - u'v)\big]\Big|_a^b.$ (Green's Identity)